

Appendix: Dynamic Programming Methods for Retirement Income

A third type of variable spending model, in addition to the methods we've discussed, uses dynamic programming methods. These methods rely on complex computing power and mathematical equations to integrate spending and asset allocation decisions more completely over the life cycle. Dynamic programming provides a road map at each point in time for optimal spending and asset allocation, which have been determined by first considering optimal future behavior stemming from today's decisions.

Because of the complexity of solving for what is optimal today based on how today's decisions will affect what happens tomorrow (and tomorrow's tomorrow, and so on), mathematical simplifications have been developed that basically involve working backward through time. Optimal end-of-life decisions are determined, which then feed into making optimal decisions at each younger age.

Due to their mathematical complexity, dynamic programming methods are mostly discussed in the realm of academia and have not yet become a common part of the toolkit for individual retirees. Gordon Irlam's AACalc (aacalc.com) and Laurence Kotlikoff's E\$Planner (esplanner.com) are two software programs based on dynamic programming that are available for household use.

Gordon Irlam and Joseph Tomlinson's 2014 article "Retirement Income Research: What Can We Learn from Economics?" in the *Journal of Retirement* tries to make dynamic programming methods more accessible to those outside of academia. It is hard to generalize about the solutions for dynamic programming methods, but Irlam and Tomlinson provide an example based on a case study in their article. In terms of understanding their findings, we can think about dynamic programming as a different way to choose an evaluation criterion. I have been discussing the choice of a PAY Rule and a visual comparison of the results as a way for individuals to decide on a retirement income strategy.